LAB 7: FUZZING FOR INPUT VALIDATION BUGS

NAME : KAVAN S **SRN : PES2UG24CS223**

SECTION : D **DATE : 11/11/2026**

1. OBJECTIVE

To identify and fix input validation bugs in a Python application using fuzz testing with the Hypothesis library.

2. SOFTWARE REQUIREMENTS

- Python 3.14
- Hypothesis
- Pytest
- Visual Studio Code

3. APPLICATION DESCRIPTION

The Spam Scoring Filter application contains three functions:

1. **sender_domain():** Extracts the domain name from an email address.
2. **first_link():** Extracts the first HTTP link from a message.
3. **shout_ratio():** Calculates the ratio of uppercase letters to total alphabetic letters.

4. FUZZ TESTING IMPLEMENTATION

Hypothesis was used to generate random strings and None values automatically.

Three test cases were created:

- test_sender_domain_no_crash()
- test_first_link_no_crash()
- test_shout_ratio_no_crash()

Command used:

python -m pytest test_spam_scoring.py -v

5. BUGS IDENTIFIED

Fuzz testing identified the following bugs:

1. **sender_domain():** Empty strings caused IndexError, and None caused AttributeError.
2. **first_link():** Missing HTTP links caused ValueError, and None caused AttributeError.

3. **shout_ratio():** Empty strings caused ZeroDivisionError, and None caused TypeError.

ORIGINAL APPLICATION OUTPUT

```
PS C:\Users\MANU\L6_L7_33_spam_scoring\lab7> python spam_scoring.py
==== Running sender_domain ====
Enter a sender email (e.g. offers@Deals.com): offers@Deals.com
Sender Domain: deals.com

==== Running first_link ====
Enter a message containing a link: Congratulations! Visit http://example.com now
First Link: http://example.com

==== Running shout_ratio ====
Enter a message (e.g. FREE prize NOW): FREE PRIZE NOW
Uppercase Ratio: 1.0
```

FUZZ TESTING FAILURES

```
test_spam_scoring.py::test_sender_domain_no_crash FAILED       [ 33%]
test_spam_scoring.py::test_first_link_no_crash FAILED          [ 66%]
test_spam_scoring.py::test_shout_ratio_no_crash FAILED         [100%]
```

```
    +----------------------------------
=================== short test summary info ====================
FAILED test_spam_scoring.py::test_sender_domain_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exce...
FAILED test_spam_scoring.py::test_first_link_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exce...
FAILED test_spam_scoring.py::test_shout_ratio_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exce...
==================== 3 failed in 15.68s ====================
```

6. BUG FIXES

The following corrections were made:

1. **sender_domain():** Added input validation and safe email domain extraction.
2. **first_link():** Used find() and added checks for missing links.
3. **shout_ratio():** Added input validation and handled division by zero.

7. FINAL TESTING

After fixing the application, the Hypothesis tests were executed again.

Command:

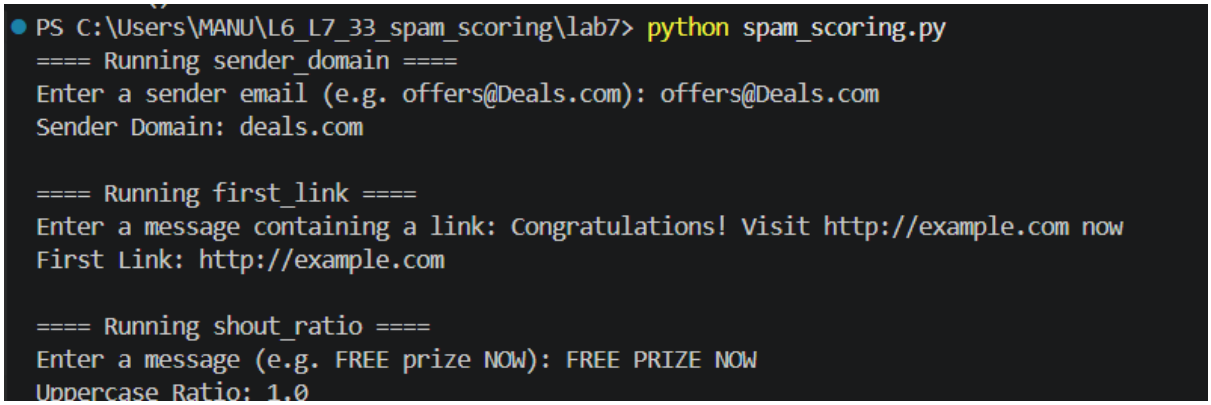
python -m pytest test_spam_scoring.py -v

Final Results:

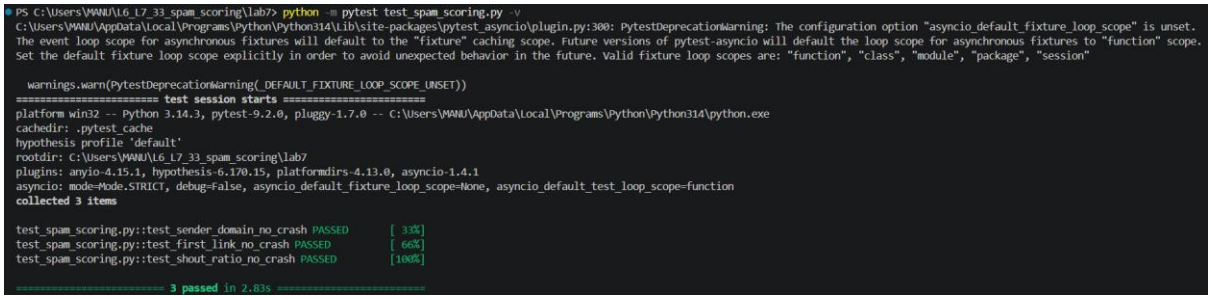
- test_sender_domain_no_crash() – PASSED
- test_first_link_no_crash() – PASSED

- test_shout_ratio_no_crash() – PASSED

FIXED APPLICATION OUTPUT



SUCCESSFUL FUZZ TESTING OUTPUT



8. CONCLUSION

Fuzz testing using Hypothesis successfully identified input validation bugs in all three functions. The bugs were fixed using defensive programming techniques. All three test cases passed after correction, improving the robustness and reliability of the application.